Answers & Solutions:

1. **B. is proportional to the distance between the plates**

2. **D. wire-wound resistor.**

3. **D. The magnetic flux in a toroid is practically all within the core**

4. **A. -150 V**

Solution:

$$E = -N\frac{d\Phi}{dt} = -200\left(\frac{30\times10^{-3}}{40\times10^{-3}}\right)$$
$$E = -150\text{ V} \rightarrow \text{Ans.}$$

5. **A. 2.5 μF**

Solution:

$$Q = CV$$
$$C = \frac{Q}{V} = \frac{6\times10^{-3}}{2.4\times10^{3}}$$
$$C = 2.5\times10^{-6}\text{ F} = 2.5\ \mu\text{F} \rightarrow \text{Ans.}$$

6. **B. $L_T = L_1 + L_2$**

7. **A. 0.32 Ω**

Solution:

$$R = \frac{\rho\ell}{A};0.08 = \frac{\rho(5)}{2\times10^{-6}}$$
$$\rho = \frac{0.08\times2\times10^{-6}}{5} = 0.032\times10^{-6}$$

If A= 1 mm^2 (i.e. half the original A)
then the length will double,
i.e. l = 2×5 = 10 m
Hence,

$$R = \frac{\rho\ell}{A} = \frac{(0.032\times10^{-6})(10)}{1\times10^{-6}}$$
$$R = 0.32\ \Omega \rightarrow \text{Ans.}$$

8. **D. Inductance**

9. **B. $C_T = C_1 + C_2 + C_3$.**

10. **D. Capacitance**

11. **A. Increases the inductance**

12. **A. An air capacitor is normally a variable type**

13. **A. 200 pF**

Solution:

If the plate area is doubled then the capacitance will double,
C = 50 pF × 2 = 100 pF
If also the plate spacing is halved, then the capacitance will again double,
C = 100 pF × 2= 200 pF → Ans.

14. **B. Resistance**

15. **D. Inductor**

16. **C. disconnect all parallel paths.**

17. **A. An electrically-controlled switch**

18. **D. 1.25 mS**

Solution:

$$Q = It$$
$$Q = CV$$
$$It = CV$$
$$t = \frac{CV}{I} = \frac{(5\times10^{-6})(500)}{2}$$
$$t = 1.25\text{ ms} \rightarrow \text{Ans.}$$

19. **C. Are comparatively nonreactive**

20. **A. 0.025 mm; 941.6 cm^2**

Solution:

$$d = \frac{1.25\text{ kV}}{50\text{ MV/m}}$$
$$d = 0.025\text{ mm}$$
$$C = \frac{\varepsilon_0\varepsilon_r A}{d}$$
$$0.2\ \mu\text{F} = \frac{8.854\times10^{-12}(6)A\left(\frac{1\text{ m}}{100\text{ cm}}\right)^2}{0.025\text{ mm}\times\frac{1\text{ m}}{1000\text{ mm}}}$$
$$A = 941.6\text{ cm}^2$$

21. **A. is always less than the smallest resistance**

22. **A. three-terminal device used to vary the voltage in a circuit.**

23. **A. dielectric**

24. **A. 144 mS**

Solution:

$$|E| = N\frac{d\Phi}{dt}$$
$$dt = \frac{Nd\Phi}{E} = \frac{(300)(12\times10^{-3})}{25}$$
$$dt = 0.144\text{ s} = 144\text{ mS} \rightarrow \text{Ans.}$$

25. **D. Capacitor**

26. **A. always less than that of the smallest capacitor**

27. **A. Compare voltages**

28. **C. 2 kV**

Solution:

$$Q = CV$$
$$V = \frac{Q}{C} = \frac{2\times10^{-6}}{1000\times10^{-12}}$$
$$V = 2000\text{ V} = 2\text{ kV} \rightarrow \text{Ans.}$$

29. **D. $R_1R_2/(R_1 + R_2)$**

30. **D. as the cross-sectional area of the coil decreases**

31. **A. 40 H**

Solution:

$$W = \frac{1}{2}LI^2$$

$$L = \frac{2W}{I^2} = \frac{(2)(80)}{(2)^2}$$
$$L = 40\ H \rightarrow \text{Ans.}$$

32. **B. 0.253**

Solution:

$$M = k\sqrt{L_1L_2}$$
$$80\ mH = k\sqrt{(250\ mH)(400\ mH)}$$
$$k = 0.253 \rightarrow \text{Ans.}$$

33. **B. 390,000 Ω**

34. **D. in a high-power, direct-current circuit**

35. **A. current is changing**

36. **A. 750 kV/m**

Solution:

Electric field strength,
$$E = \frac{V}{d} = \frac{30}{0.04\times10^{-3}}$$
$$E = 750\ kV/m \rightarrow \text{Ans.}$$

37. **A. 10 μF; 2.4 μF**

Solution:

$$C_{eq(parallel)} = 6\mu F + 4\mu F$$
$$C_{eq(parallel)} = 10\ \mu F \rightarrow \text{Ans.}$$
$$C_{eq(series)} = \frac{C_1C_2}{C_1+C_2} = \frac{6\mu F(4\mu F)}{6\mu F + 4\mu F}$$
$$C_{eq(series)} = 2.4\ \mu F \rightarrow \text{Ans.}$$

38. **C. surface-mount resistors.**

39. **A. An air capacitor is normally a variable type**

40. **C. decreases with an increase in operating temperature.**

41. **C. 5 mH, 0.25**

Solution:

$$L_{eq} = L_1 + L_2 + 2M$$
$$60\ mH = 40\ mH + 10\ mH + 2M$$
$$M = 5\ mH \rightarrow \text{Ans.}$$
$$M = k\sqrt{L_1L_2}$$
$$5\ mH = k\sqrt{(40\ mH)(10\ mH)}$$
$$k = 0.25 \rightarrow \text{Ans.}$$

42. **A. charge to p.d. between plates**

43. **C. the current is changing**

44. **C. 25 V in the opposite direction to the applied voltage**

45. **D. metal-film resistors.**

46. **B. 0.026 μΩ-m**

Solution:

$$R = \frac{\rho\ell}{A}$$
$$\rho = \frac{RA}{\ell} = \frac{(5\ \Omega)(2.6\times10^{-6}\ m^2)}{500\ m}$$
$$\rho = 26\times10^{-9}\Omega\ m = 0.026\ \mu\Omega\text{-m} \rightarrow \text{Ans.}$$

47. **D. 84.61 Ω**

Solution:

$$R = R_o(1 + \propto \Delta T)$$
$$R = 75\Omega\left[1 + \left(0.00427/^{o}C\right)(30-0)\right]$$
$$R = 84.61^{o}C \rightarrow \text{Ans.}$$

48. **C. There is no correlation between the physical size of a resistor and its resistance value.**

49. **A. bringing the metal plates closer together**

50. **A. 2.5 mC**

Solution:

$$Q = CV = (10\times10^{-6})(250)$$
$$Q = 2.5\times10^{-3}\ C = 2.5\ mC \rightarrow \text{Ans.}$$

END

END